Laboratory Exercise 10 – External Flow Over a Cylinder

Experiment Data Sheet

Name: Grant Keller Date: _____

Names of Lab Partners (for reference):

Tim Strader	Peter Fernandez

Environmental Conditions:

Barometric Pressure, P_{atm}	Barometric Pressure Uncertainty, U_{P_atm}	Room Temperature, T_{atm}	Room Temperature Uncertainty, U_{T_atm}
99.7 mbar	± 0.1 mbar	21 °C	± 1 °C

Water Density:

Water Volume, Vol_{water}	Water Volume Uncertainty, U_{Vol_water}	Water Mass, m_{water}	Water Mass Uncertainty, U_{m_water}
24 mL	± 0.5 mL	24.2 g	± 0.25 g

Geometry:

Cylinder Diameter, D	Uncertainty of Cylinder Diameter, U_D	Cylinder Length, $L_{cylinder}$	Uncertainty of Cylinder Length, $U_{L_cylinder}$
24.9 mm	0.1 mm	5 7/8 in	1/32 in

Water channel width, $w_{channel}$	Uncertainty of water channel width, $U_{w_channel}$	Wetted height of water channel, h_{wetted}	Uncertainty of water channel wetted height, U_{h_wetted}
6 1/4 in	1/32 in	5 1/8 in	1/32 in

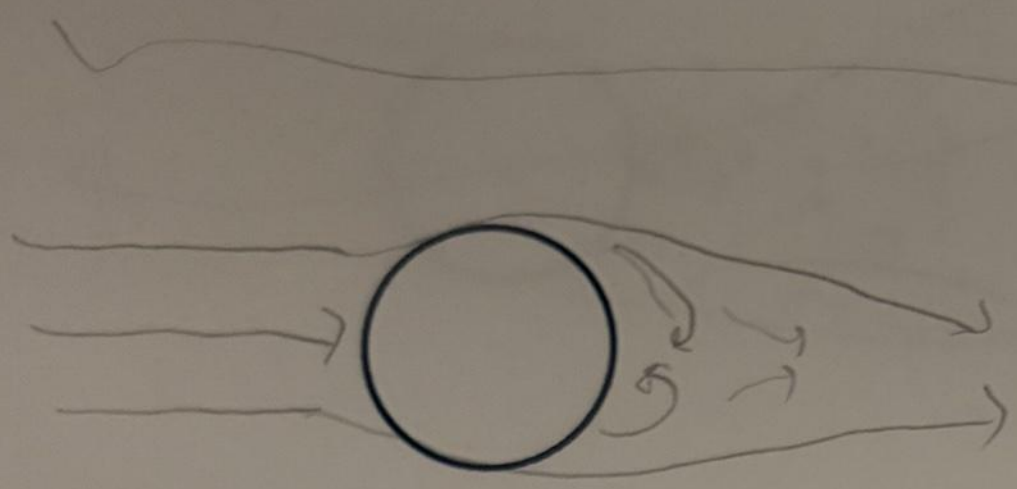
L_1 (see diagram)	Uncertainty of L_1	L_2 (see diagram)	Uncertainty of L_2
2 1/16 in		6 1/4 in	1/32 in

Drag Force and Flow Pattern Observation:

Motor Controller Setpoint = 10:

	Parameter	Uncertainty
Flowrate, *Q*	52,7 G/min	± 0,5 G/min
Tension Force, *T*	0,196 N	0,01 N

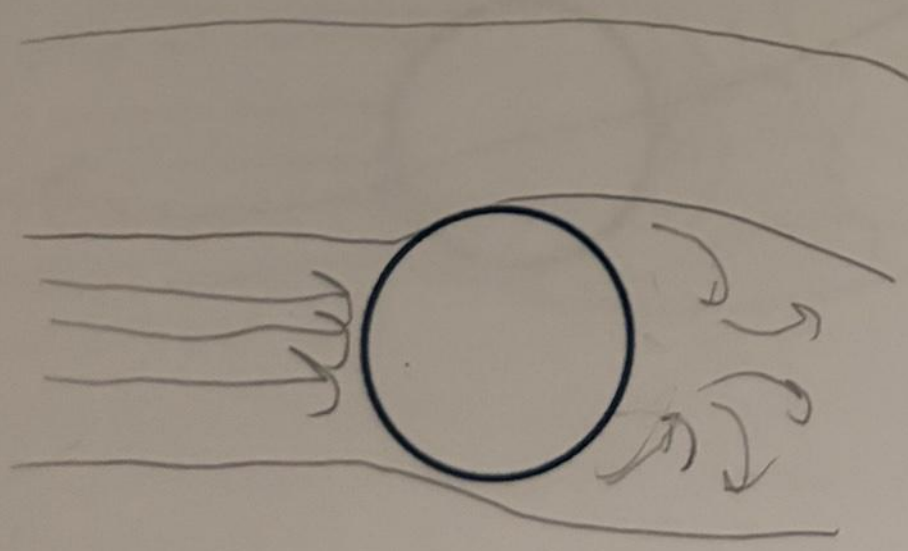
Sketch of Flow Pattern, with Approximate Flow Separation Angle:



Motor Controller Setpoint = 20:

	Parameter	Uncertainty
Flowrate, *Q*	106,2	± 0,5 G/min
Tension Force, *T*	1,02 N	±0,01 N

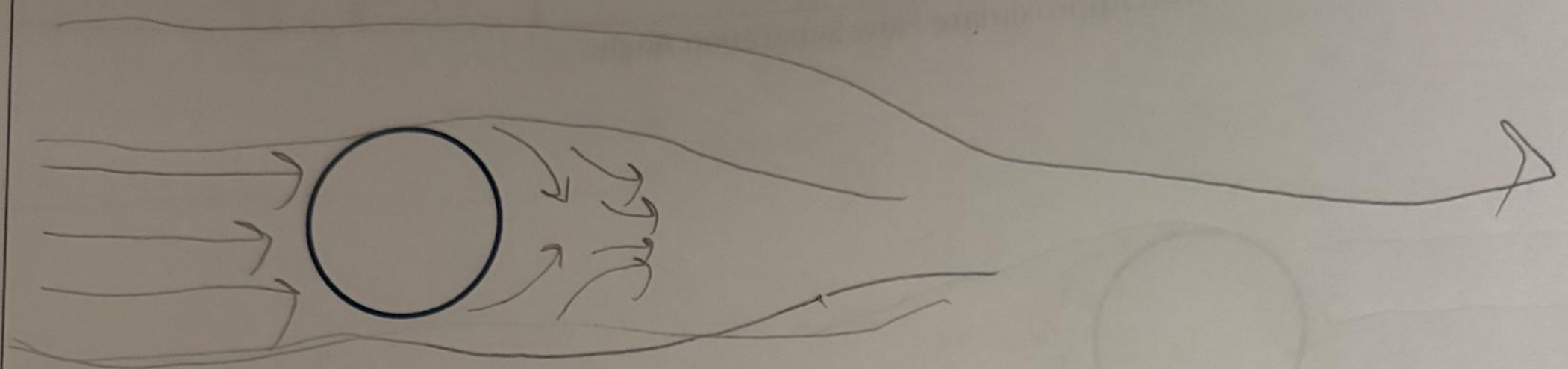
Sketch of Flow Pattern, with Approximate Flow Separation Angle:



Motor Controller Setpoint = 30:

	Parameter	Uncertainty
Flowrate, *Q*	162,8	± 1,2 g/min
Tension Force, *T*	2,86	± 1,1 N

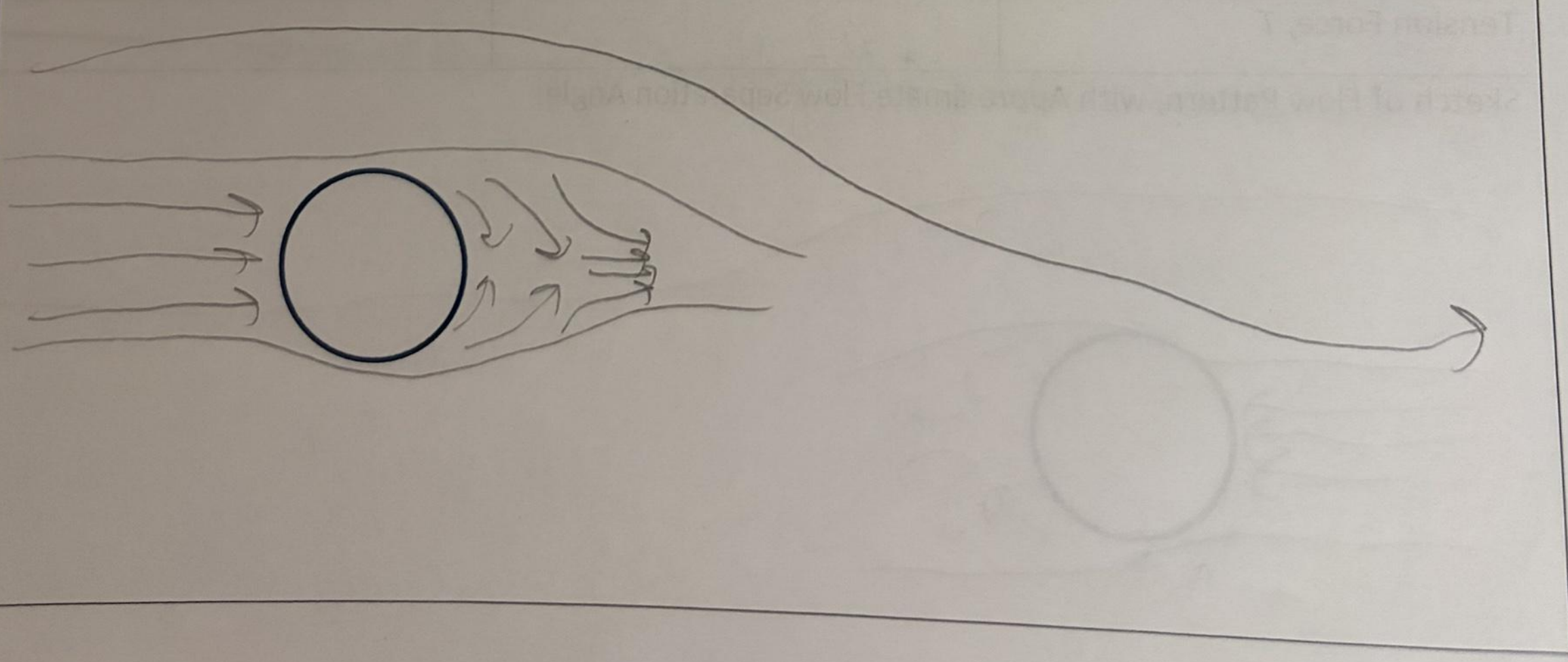
Sketch of Flow Pattern, with Approximate Flow Separation Angle:



Motor Controller Setpoint = 40:

	Parameter	Uncertainty
Flowrate, *Q*	208,2 g/min	± 3,5 g/min
Tension Force, *T*	3,68 N	± 1,1 N

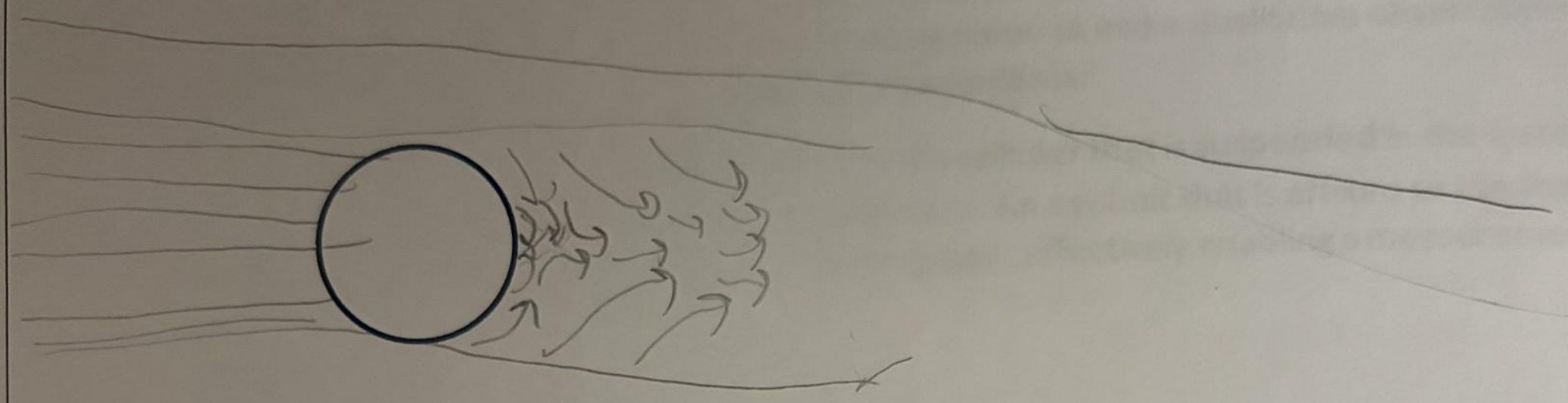
Sketch of Flow Pattern, with Approximate Flow Separation Angle:



Motor Controller Setpoint = 50:

	Parameter	Uncertainty
Flowrate, *Q*	223,5 G/min	± 5,5 G/min
Tension Force, *T*	3,98 N	± 0,1 N

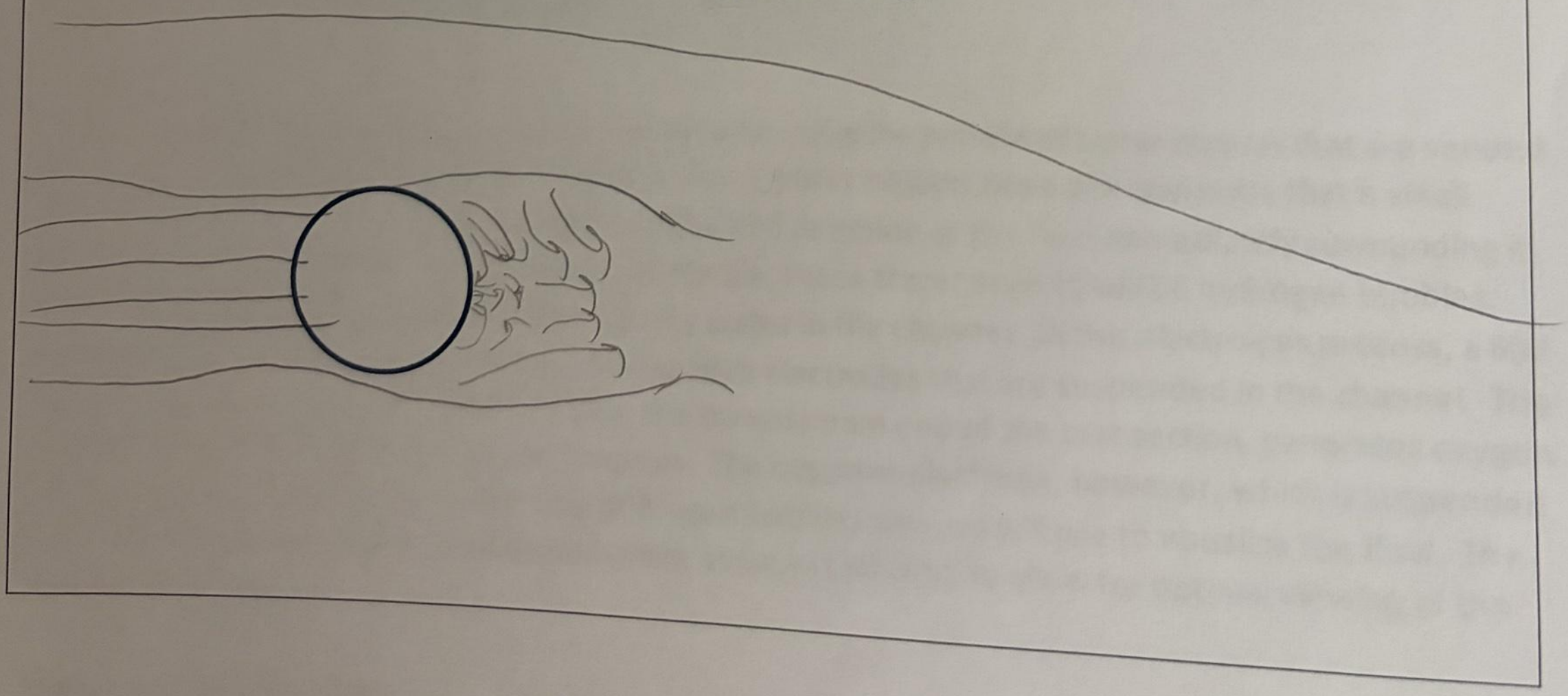
Sketch of Flow Pattern, with Approximate Flow Separation Angle:



Motor Controller Setpoint = 60:

	Parameter	Uncertainty
Flowrate, *Q*	247,5 G/min	± 7 G/min
Tension Force, *T*	4,97 G/min	± 0,12 N

Sketch of Flow Pattern, with Approximate Flow Separation Angle:



Instructor Signature: _____